DESIGN TO COST · PART COST ESTIMATOR

Work instruction with a worked example

From the 3D model to the tool drawings and the PO pack: the complete design loop, step by step. Version 2026-10-26.

Example: L mounting bracket EX26-0001, 2 mm galvanised steel, 40 mm wide, legs 250 and 80 mm, 6 holes. Rev A has a 0.5 mm bend radius (the simulation asks for a correction), Rev B has 2 mm. Also a plastic cover EX26-0002 with its mould.

Sample files (Help page → Work instruction): example_bracket_revA.step, example_bracket_revB.step, plastic_cover.step, and every output of this example.

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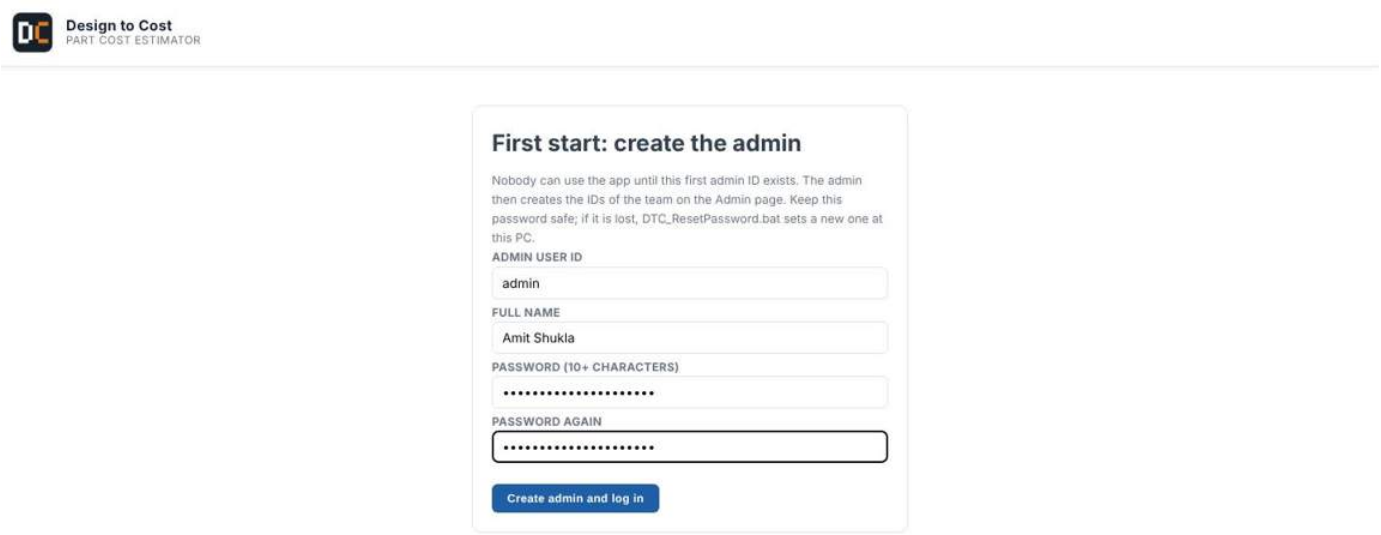
1 · Start

Install / update (Windows): the update kit in the base folder (e.g. D:\DTC). **DTC_Update.bat** makes a backup, installs the new version and merges your master data; **DTC_Start.bat** starts the app and opens the browser. **DTC_Backup.bat**, **DTC_Restore.bat**, **DTC_Rollback.bat**, **DTC_ResetPassword.bat** do what their names say.
Menu: Dashboard (projects) · Parts (register, part panel) · Suppliers · Import (STEP / Excel) · Harness & PCB · Estimate (quick costing of a STEP without registering) · Rates (approved rates by country, review) · Master data (rules, machines, steels, bases, formulas) · Help · Admin (users, permissions, activity, backups).
Rules of the software: no number is hard-coded (everything comes from master data or the rate library); every value can be clicked to see how it is made; only approved rates are used; stored cost versions never change by themselves; EXAMPLE values are flagged in red until you replace them.

2 · The worked example, step by step

Step 1 · First start: create the admin

• Start the app with DTC_Start.bat; the browser opens http://127.0.0.1:8000.
• With no user yet the app shows “First start”: enter the admin ID, full name and a password (10+ characters).
• Keep the password safe. A forgotten password is reset at the PC with DTC_ResetPassword.bat.
Output: Admin login.

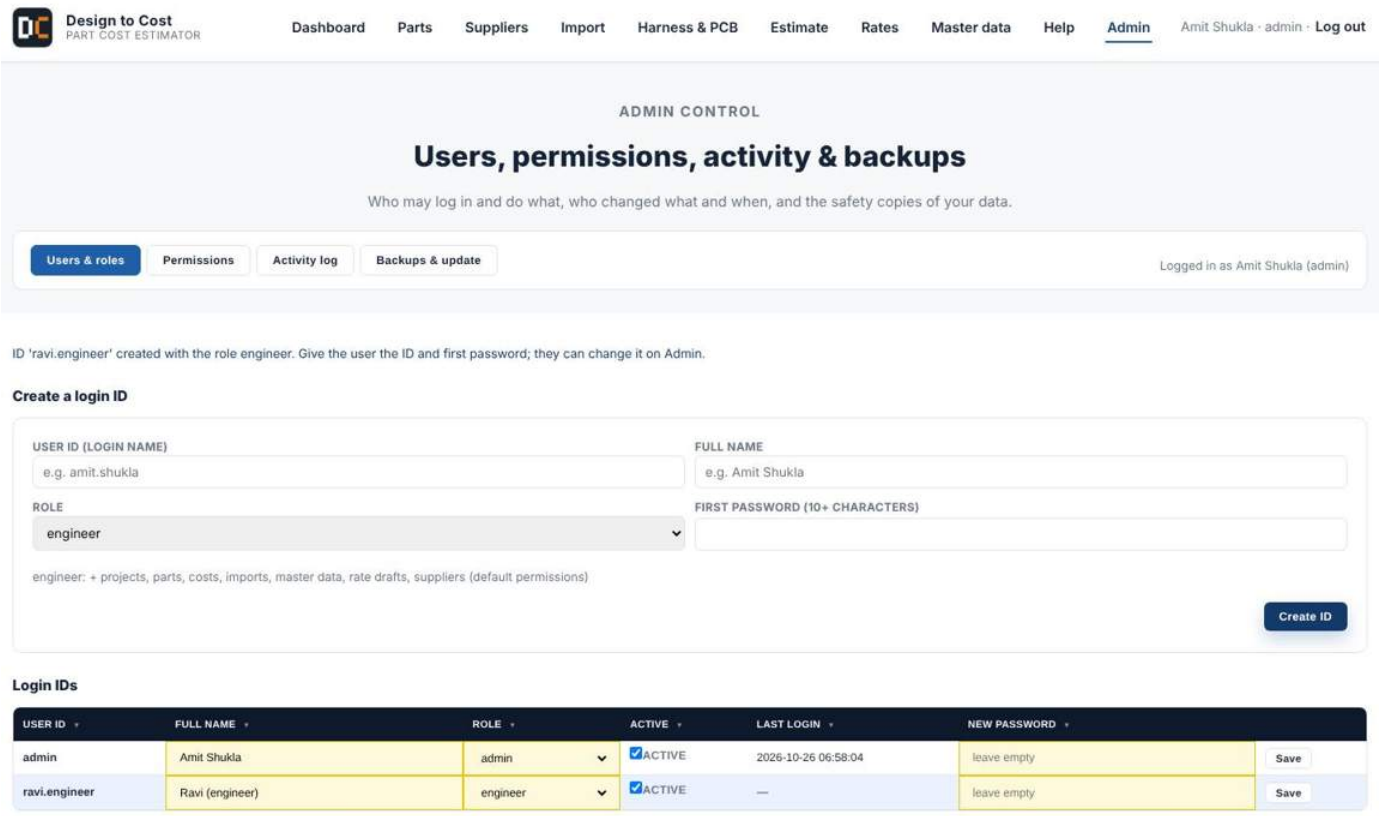


First start: create the admin ID (only once).

Step 2 · Create the team's login IDs

- Admin (menu, top right) → Users & roles → Create a login ID: user ID, name, role, first password.
- Roles: viewer (reads), engineer (changes data), approver (+ approves rates), admin (+ users, permissions, backups).
- Admin → Permissions sets the lowest role for each action.

Output: One login per person; every change is recorded under it.

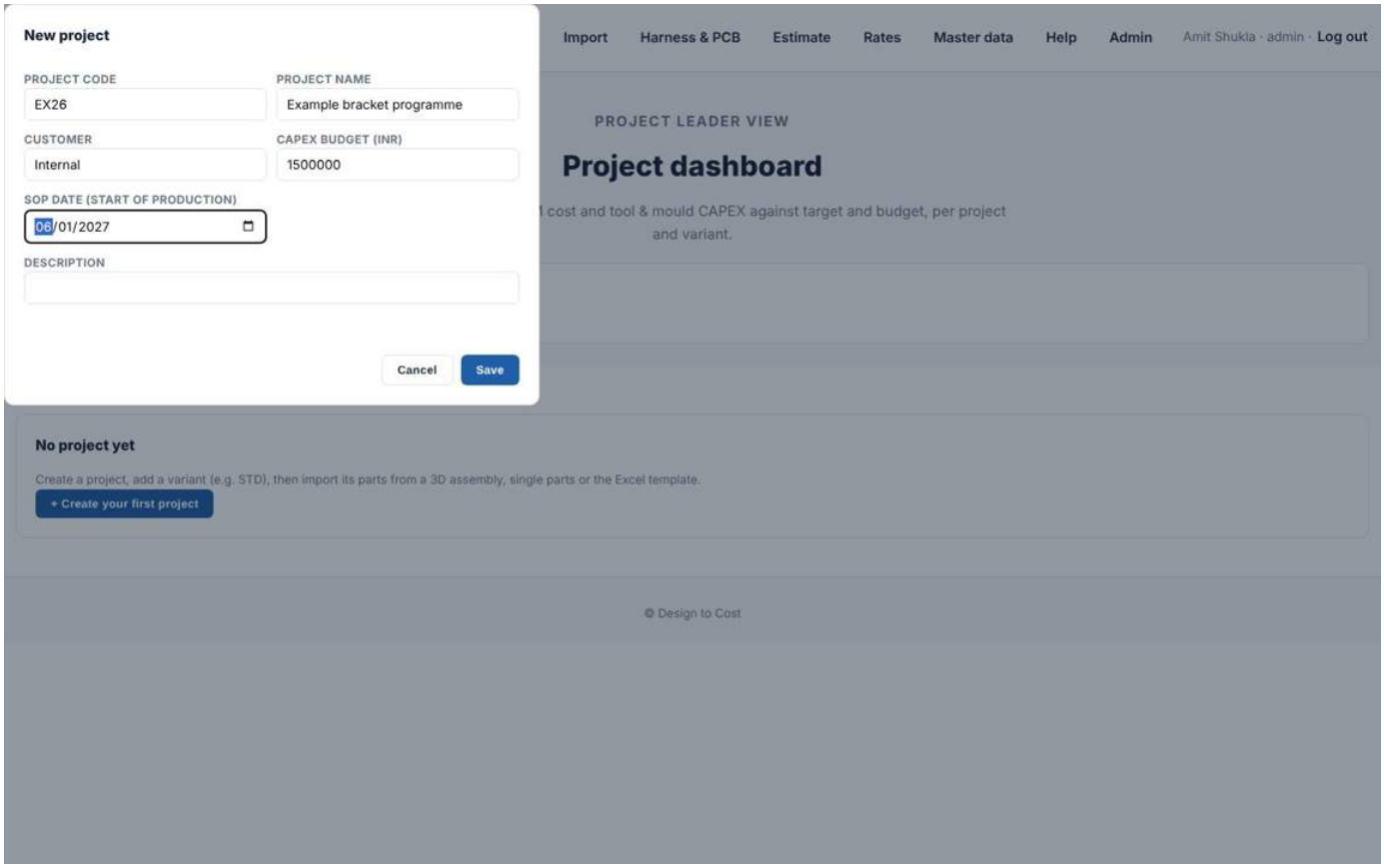


Admin → Users & roles: create a login ID for each team member.

Step 3 · Create the project and its variant

• Dashboard → + New project: code, name, customer, CAPEX budget, SOP date → Save.
• + Variant: code (e.g. STD), name, target BOM cost, annual volume → Save.

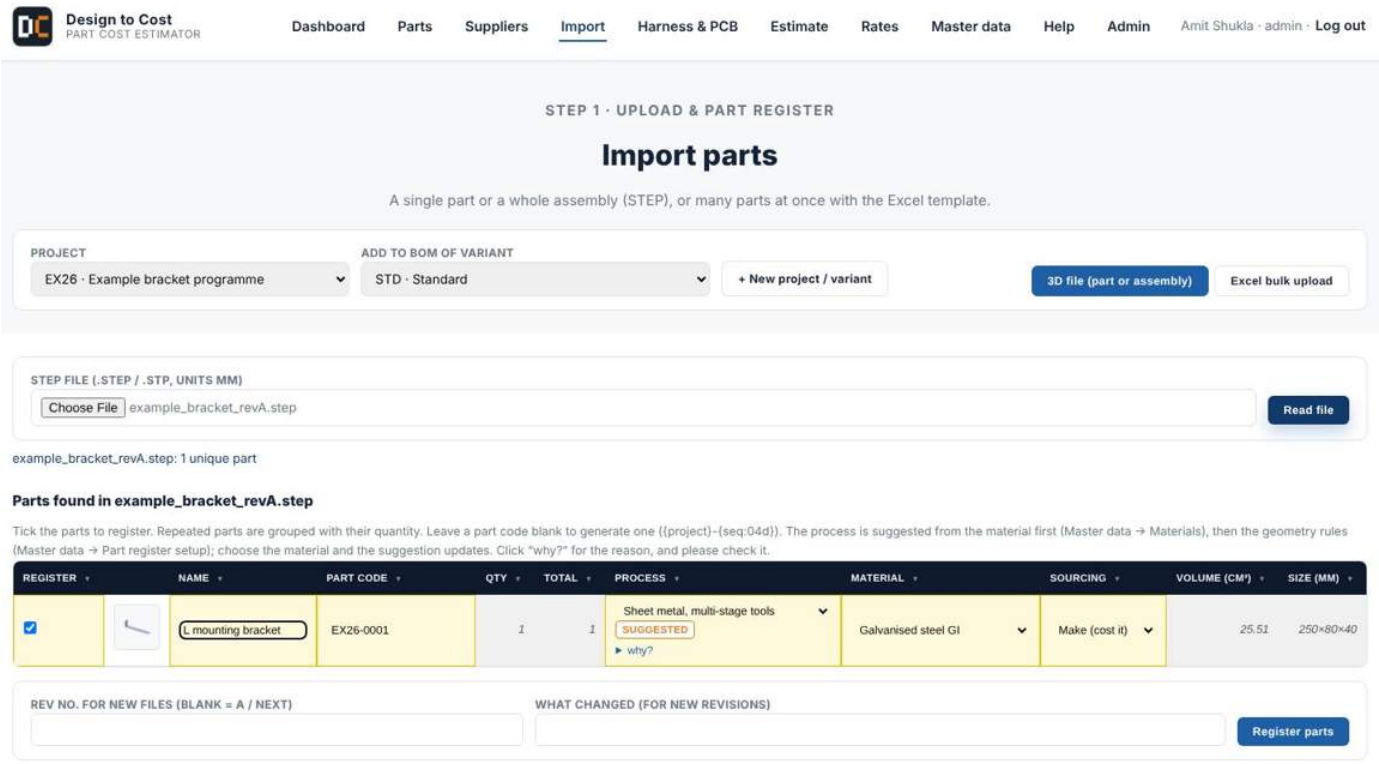
Output: Project EX26 with variant STD (200,000 parts / year).



Dashboard → + New project: code, name, customer, CAPEX budget, SOP date.

Step 4 · Import the 3D model (Rev A)

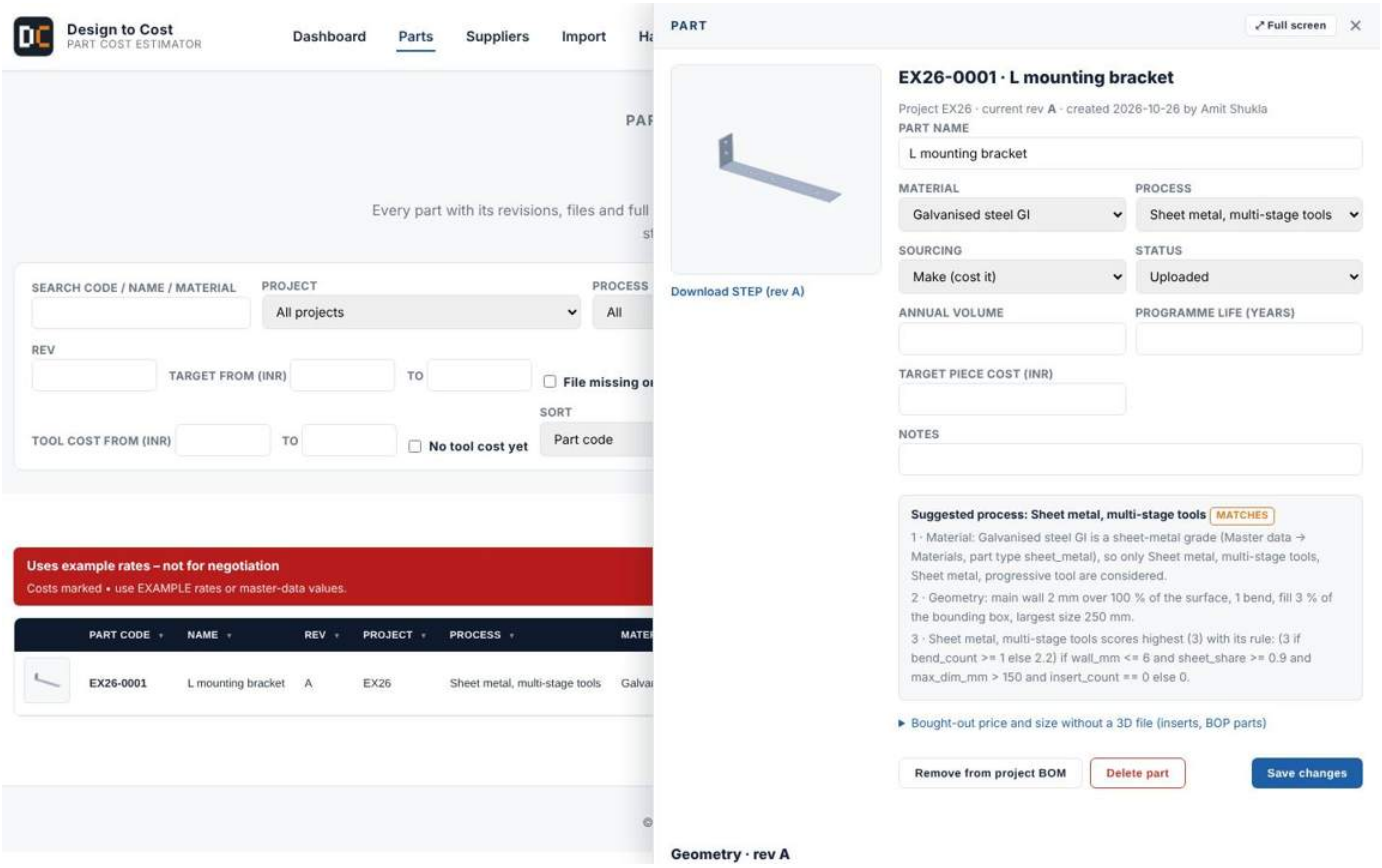
• Import → 3D file → choose the STEP file → Read file (sample: example_bracket_revA.step).
• Choose the material (Galvanised steel GI); the process is suggested with its reason (why?). Give the part code and name.
• Register parts. The geometry, net weight, RM cost and tool cost are worked out automatically.
Output: Part EX26-0001 rev A in the register and in the BOM of STD.



Import → STEP file: read the 3D model, choose the material; the process is suggested with its reason.

Step 5 · Find the part in the register

• Parts → filter by project, process, status; ■ in a column header filters any column.
• Click a part to open its panel; ■ Full screen gives the whole window.

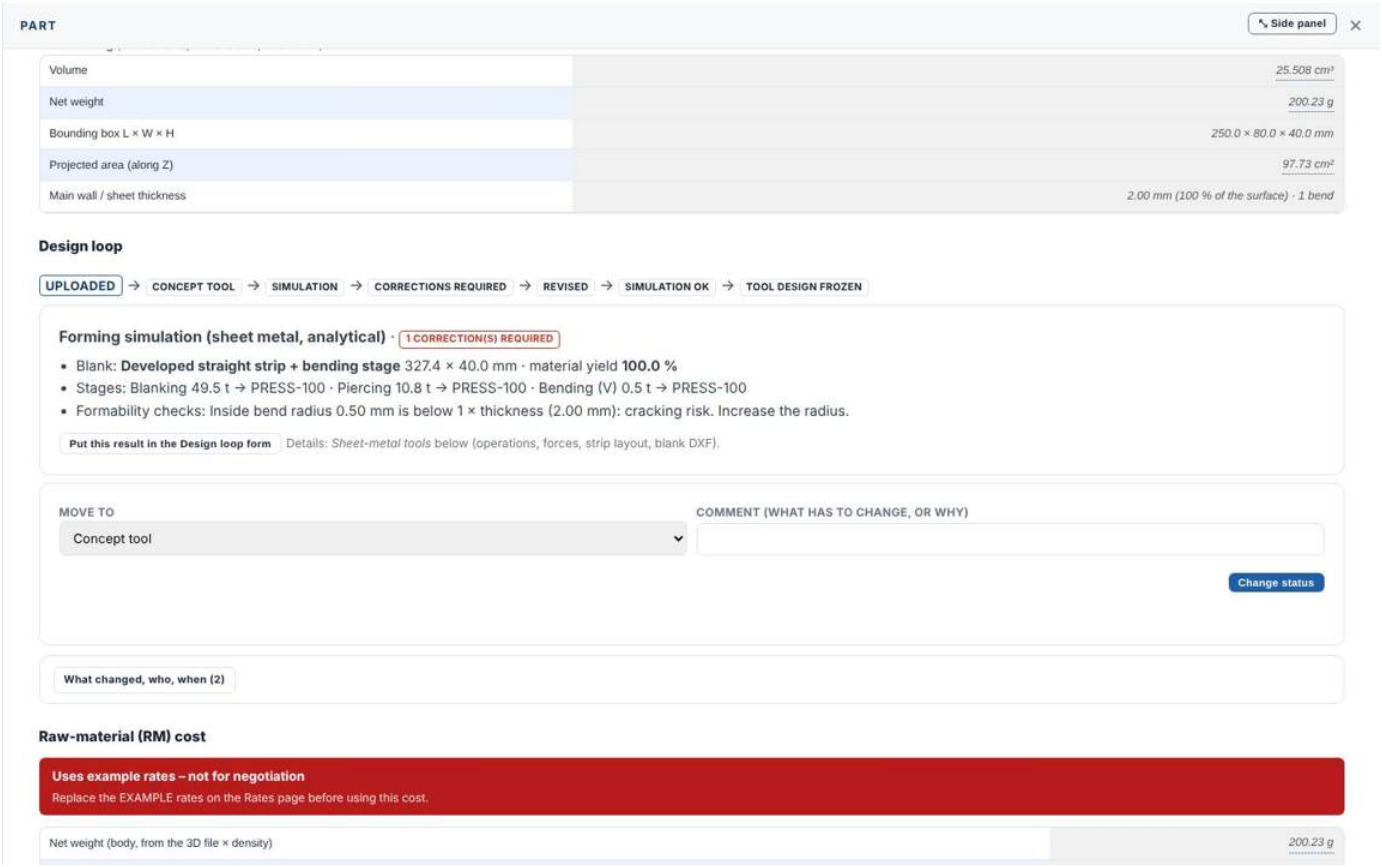


Parts: the register with status, process, material, volume, RM and tool cost.

Step 6 · Check the part and download its drawing

- Check name, material, process, sourcing, volume; Save changes if needed.
- Geometry from the 3D model: volume, net weight, size, thickness, bends.
- Part drawing PDF / DXF: three views, overall dimensions, title block.

Output: EX26-0001_part_drawing_revA.pdf

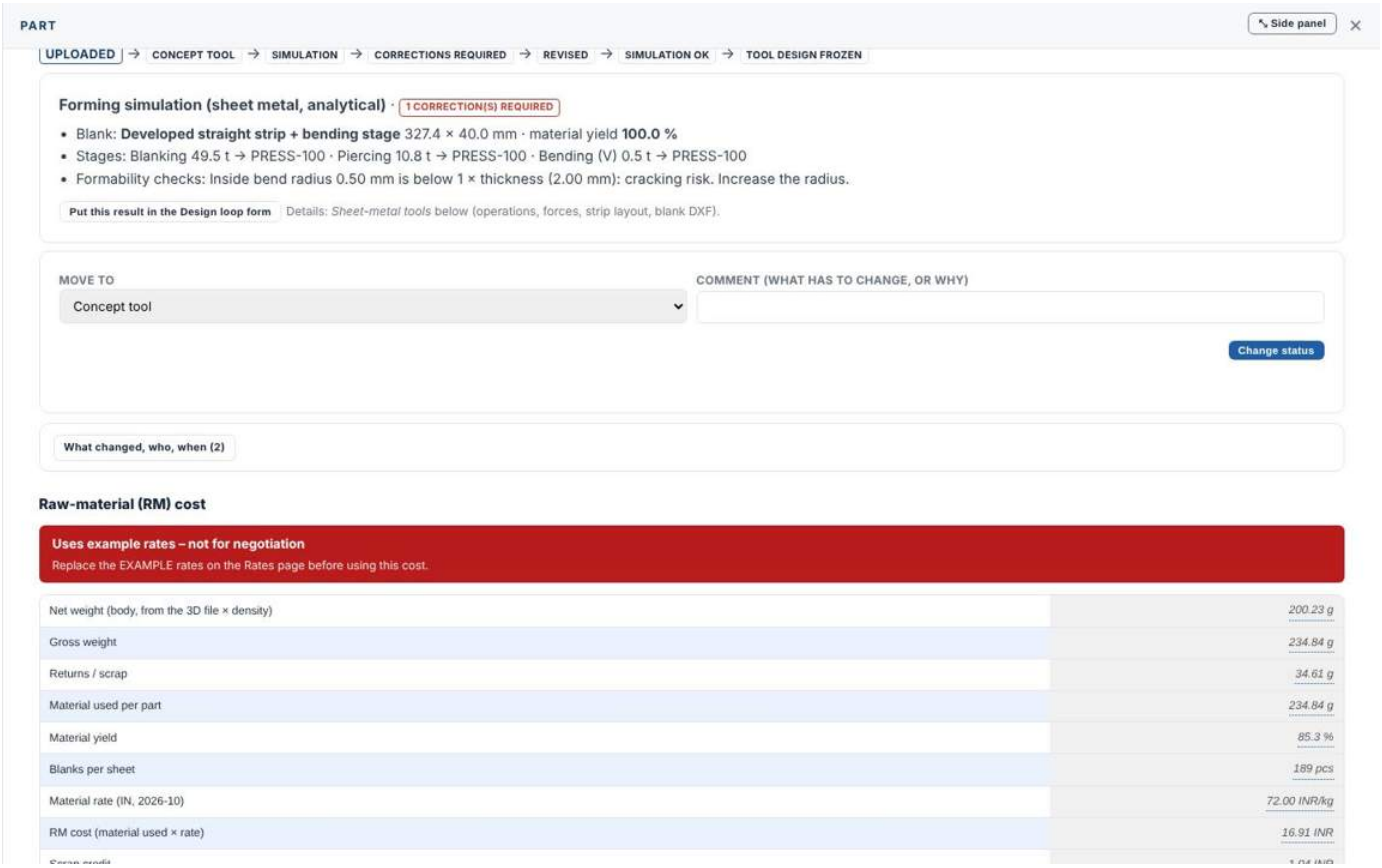


The part: details, suggested process, geometry from the 3D model and the Part drawing (PDF · DXF).

Step 7 · Design loop: start

• The Design loop card shows the stages and the simulation result of the current model.
• Move to → Concept tool → Change status (the software proposes the tool concept).

Output: Status Concept tool.

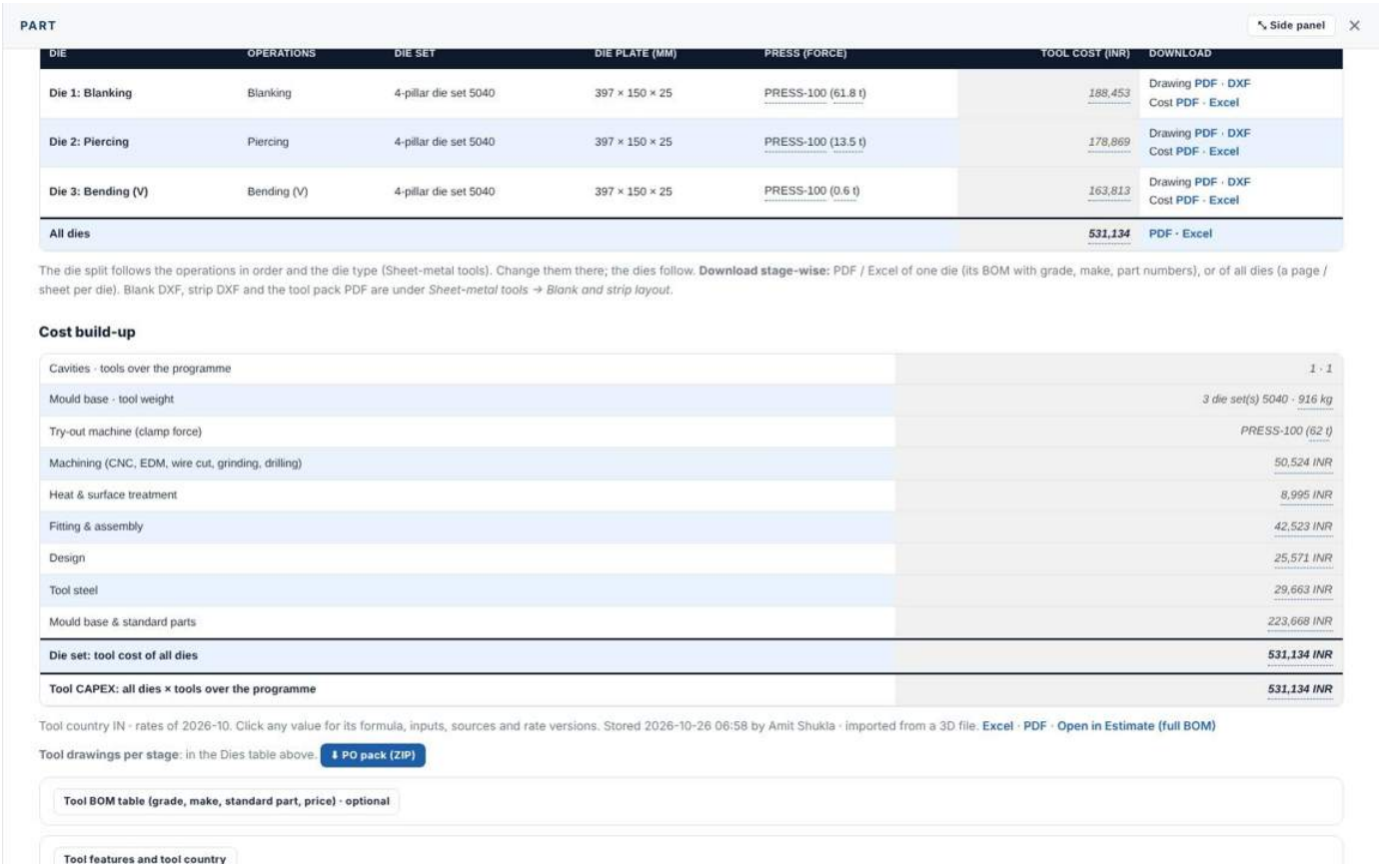


Design loop at Uploaded: the forming simulation result is already shown (Rev A: bend radius too tight).

Step 8 · Stage tools: the dies suggested

- Tool cost (mould / die): one die per operation (Blanking, Piercing, Bending) or one progressive die.
- Each die: die set, die plate L × W × H, press for its force, cost. Click any value for its traceability.
- Download per die: tool drawing PDF / DXF and cost PDF / Excel; All dies: one page per stage.

Output: EX26-0001_stage1…3_tool_drawing.pdf / .dxf, …_tool_cost.pdf



PART

Side panel

DIE	OPERATIONS	DIE SET	DIE PLATE (MM)	PRESS (FORCE)	TOOL COST (INR)	DOWNLOAD
Die 1: Blanking	Blanking	4-pillar die set 5040	397 × 150 × 25	PRESS-100 (61.8 t)	188,453	Drawing PDF · DXF Cost PDF · Excel
Die 2: Piercing	Piercing	4-pillar die set 5040	397 × 150 × 25	PRESS-100 (13.5 t)	178,869	Drawing PDF · DXF Cost PDF · Excel
Die 3: Bending (V)	Bending (V)	4-pillar die set 5040	397 × 150 × 25	PRESS-100 (0.6 t)	163,813	Drawing PDF · DXF Cost PDF · Excel
All dies					531,134	PDF · Excel

The die split follows the operations in order and the die type (Sheet-metal tools). Change them there; the dies follow. **Download stage-wise:** PDF / Excel of one die (its BOM with grade, make, part numbers), or of all dies (a page / sheet per die). Blank DXF, strip DXF and the tool pack PDF are under *Sheet-metal tools → Blank and strip layout*.

Cost build-up

Cavities · tools over the programme	1 · 1
Mould base · tool weight	3 die set(s) 5040 · 916 kg
Try-out machine (clamp force)	PRESS-100 (62 t)
Machining (CNC, EDM, wire cut, grinding, drilling)	50,524 INR
Heat & surface treatment	8,995 INR
Fitting & assembly	42,523 INR
Design	25,571 INR
Tool steel	29,663 INR
Mould base & standard parts	223,668 INR
Die set: tool cost of all dies	**531,134 INR**
Tool CAPEX: all dies × tools over the programme	**531,134 INR**

Tool country IN · rates of 2026-10. Click any value for its formula, inputs, sources and rate versions. Stored 2026-10-26 06:58 by Amit Shukla · imported from a 3D file. Excel · PDF · Open in Estimate (full BOM)

Tool drawings per stage: in the Dies table above. ⬇ PO pack (ZIP)

Tool BOM table (grade, make, standard part, price) · optional

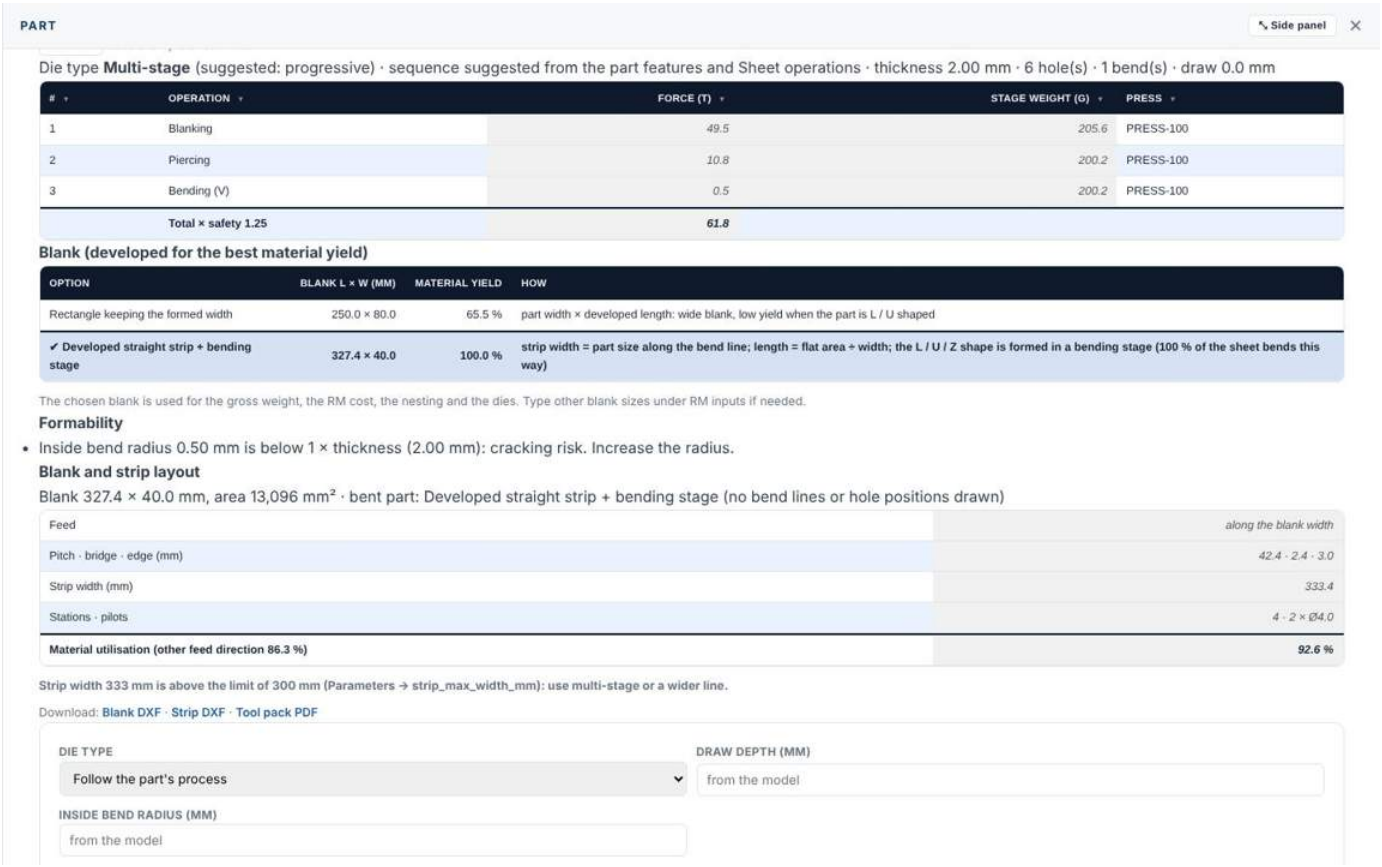
Tool features and tool country

Tool cost: the dies suggested stage-wise (die set, die plate, press, cost) with drawing and cost downloads per die.

Step 9 · Sheet-metal tools and the blank

- Operations in order (↑ ↓ ✕, + Add, Reset to suggested), die type, forces, presses.
- Blank developed for the best material yield (L shape → straight strip + bending stage).
- Blank DXF, strip DXF and tool pack PDF under Blank and strip layout.

Output: EX26-0001_blank.dxf, EX26-0001_tool_pack.pdf

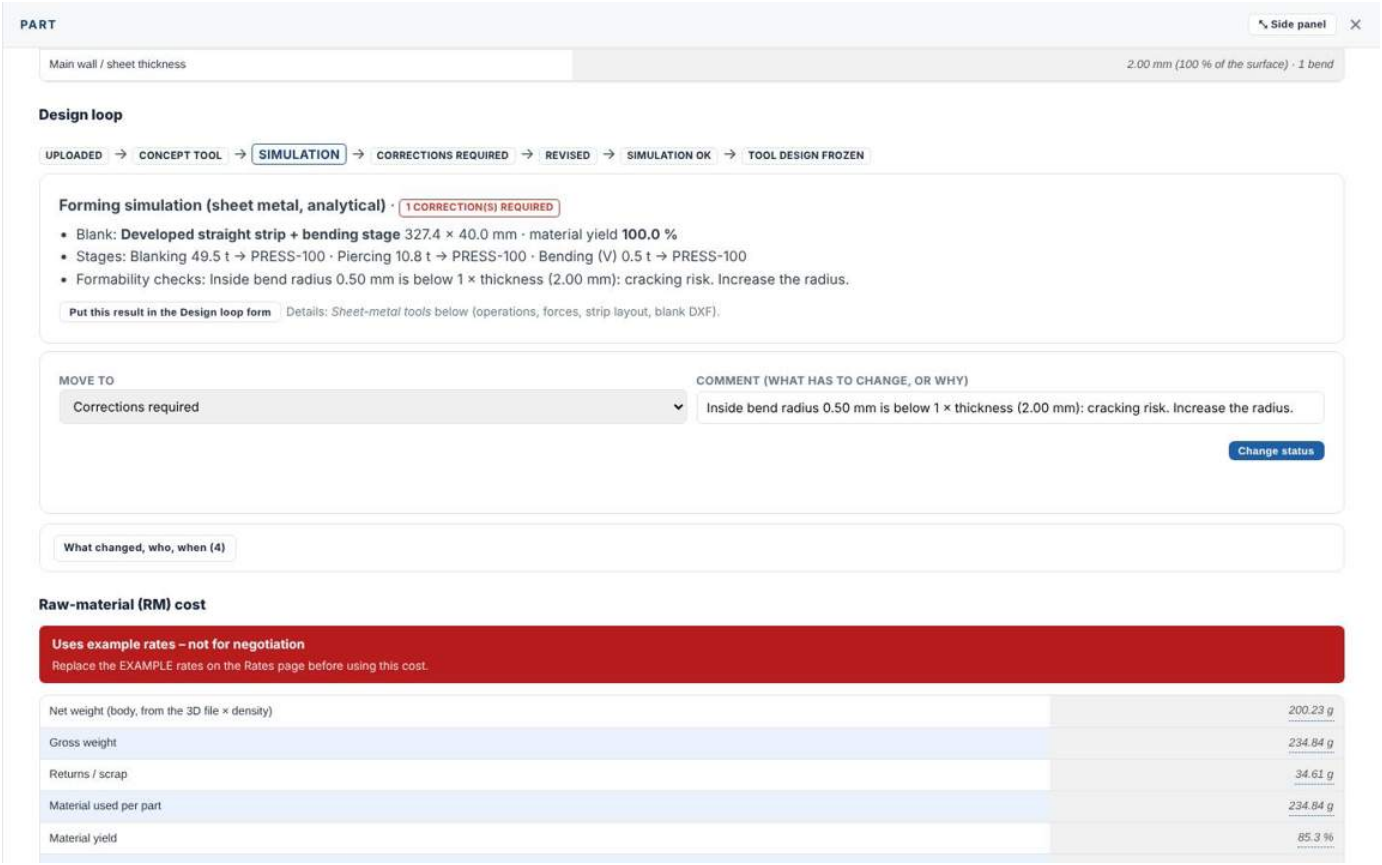


Sheet-metal tools: operations in order with forces and presses, and the developed blank with its material yield.

Step 10 · Design loop: simulation and correction

• Move to → Simulation → Change status.
• Forming simulation: 1 correction required (bend radius 0.5 mm < 1 × thickness).
• Put this result in the Design loop form → Corrections required with the comment → Change status.
• Send the comment to the product designer for a corrected model.

Output: Status Corrections required, comment in the timeline.

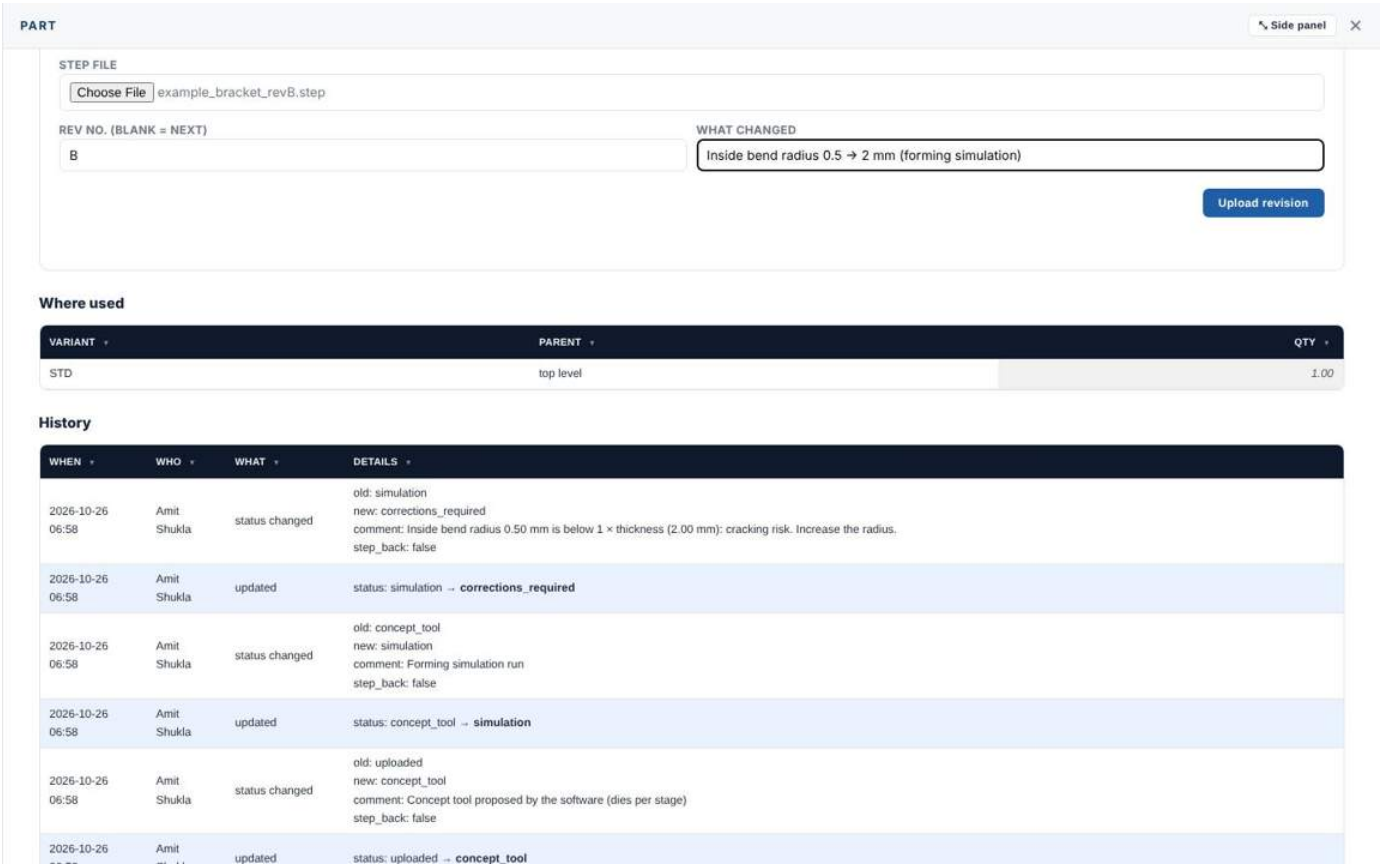


Simulation: 1 correction required → “Put this result in the Design loop form” fills Corrections required and the comment.

Step 11 · Upload the corrected model (Rev B)

• Part panel → Upload a new revision: STEP file, Rev no. B, what changed → Upload revision.
• Rev A, its files and costs stay; the status moves to Revised by itself and all costs are recalculated.

Output: Rev B, status Revised.

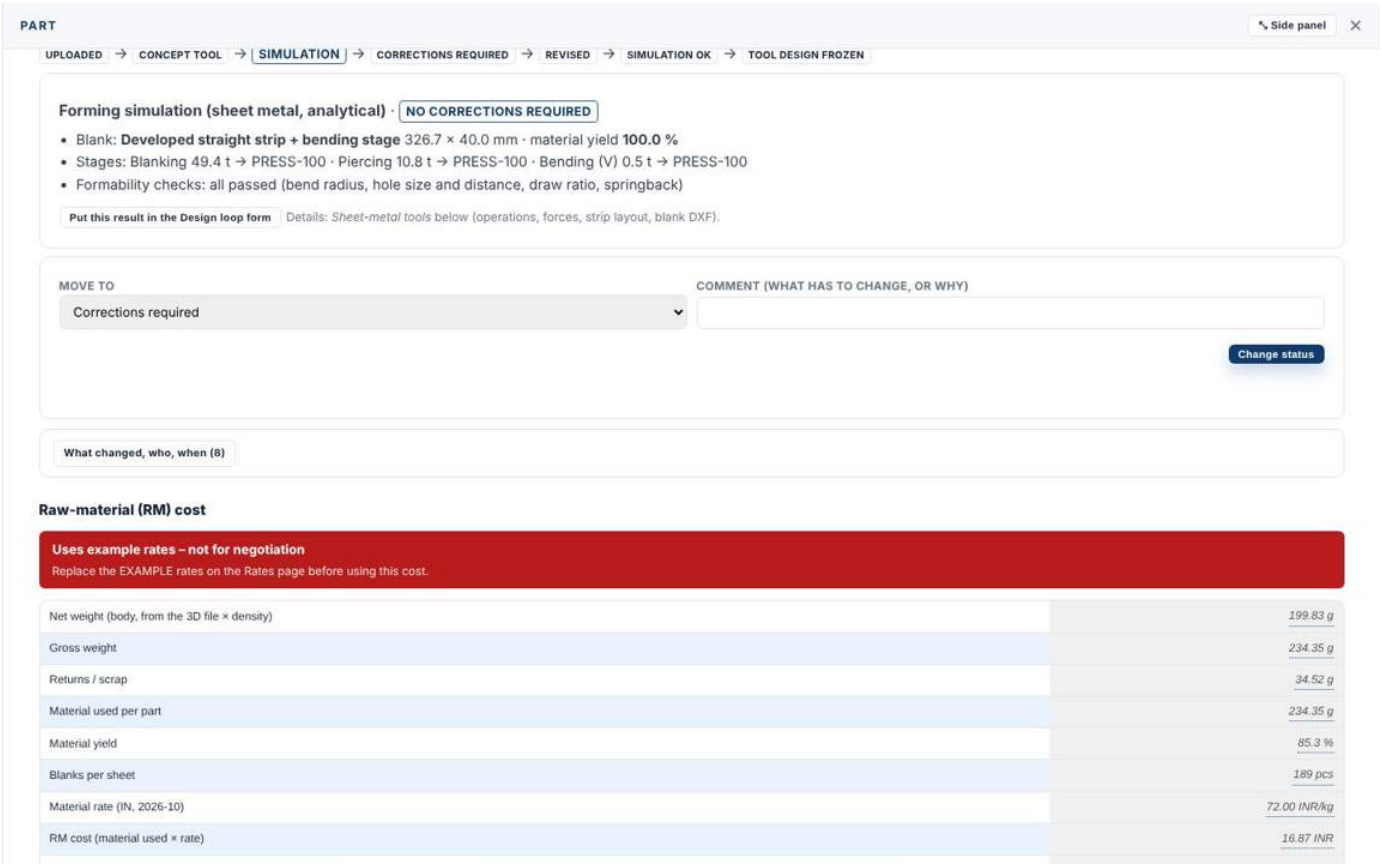


Upload the corrected model as Rev B with what changed; the status moves to Revised by itself.

Step 12 · Design loop: simulation OK

• Move to → Simulation → Change status: No corrections required.
• Put this result in the Design loop form → Simulation OK → Change status.

Output: Status Simulation OK.

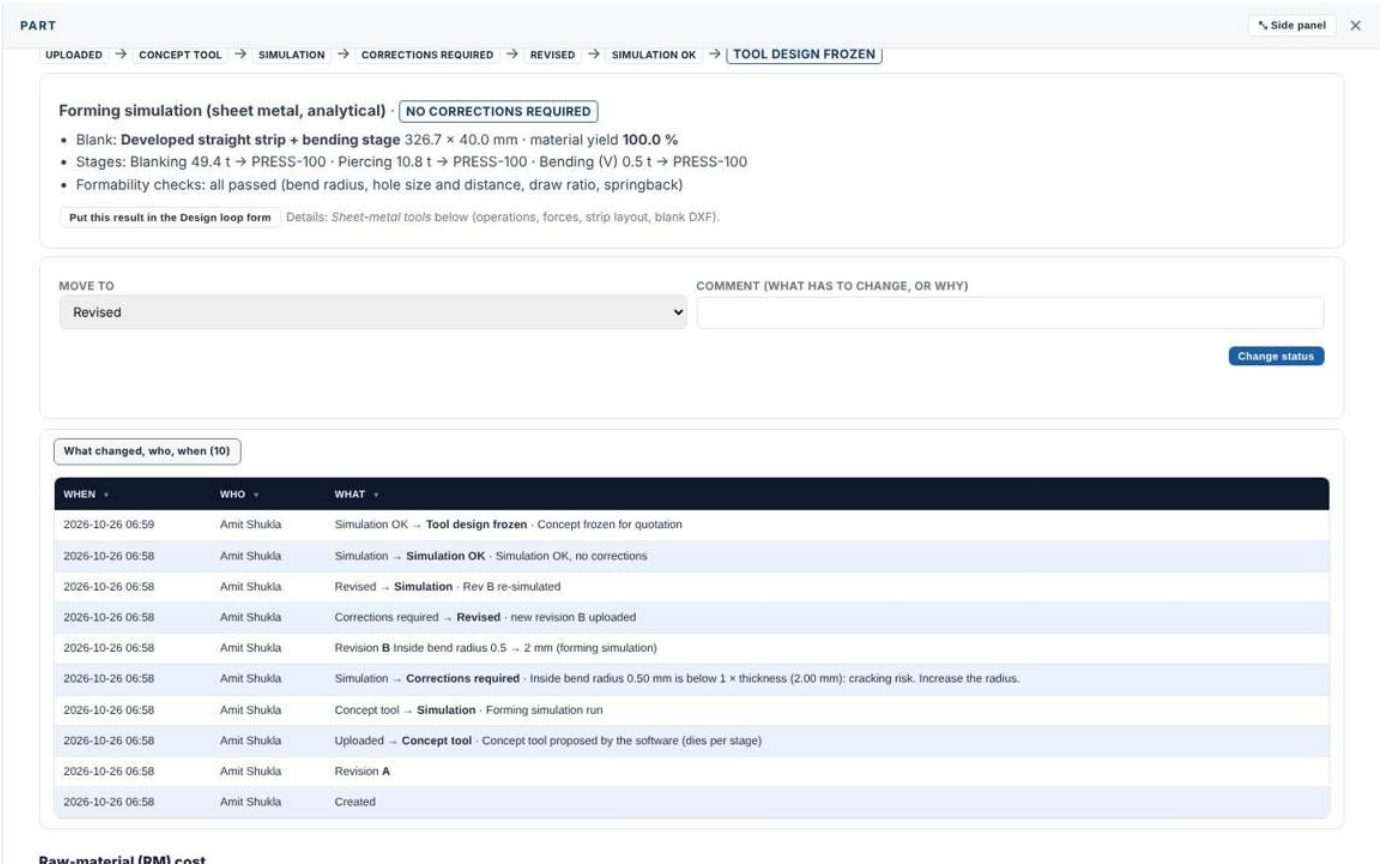


Rev B: No corrections required → Simulation OK, then Tool design frozen.

Step 13 · Design loop: tool design frozen

• Move to → Tool design frozen (comment: concept frozen for quotation) → Change status.
• What changed, who, when: every move, revision and comment.

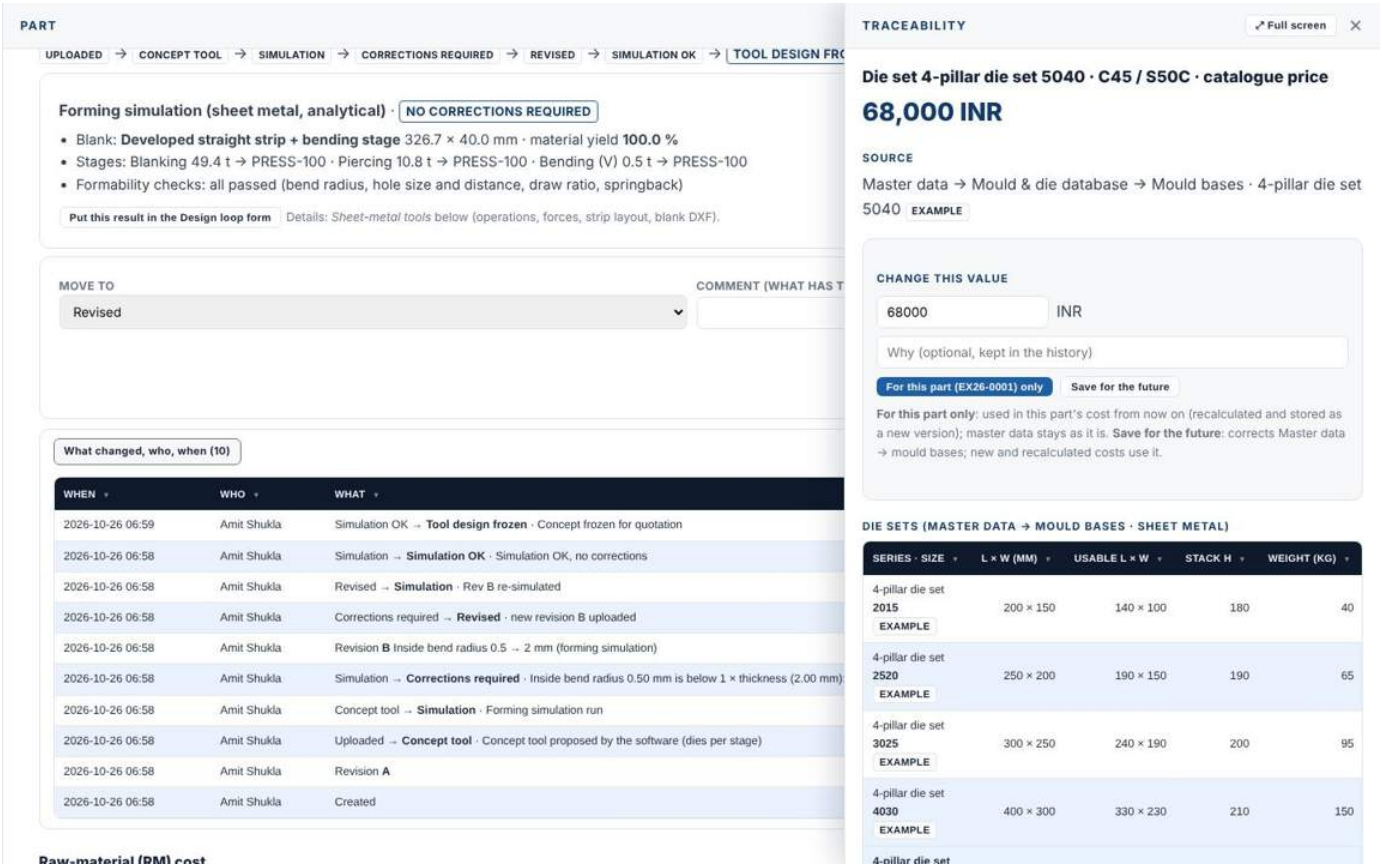
Output: Status Tool design frozen: ready for RFQ / PO.



Tool design frozen: the timeline keeps every status change and revision with who, when and why.

Step 14 · Traceability: die set / mould base

- Click the die set or mould base price: every base of that type with size, weight, price.
- Use for this part, or + Add a row (saved to master data, then selectable).

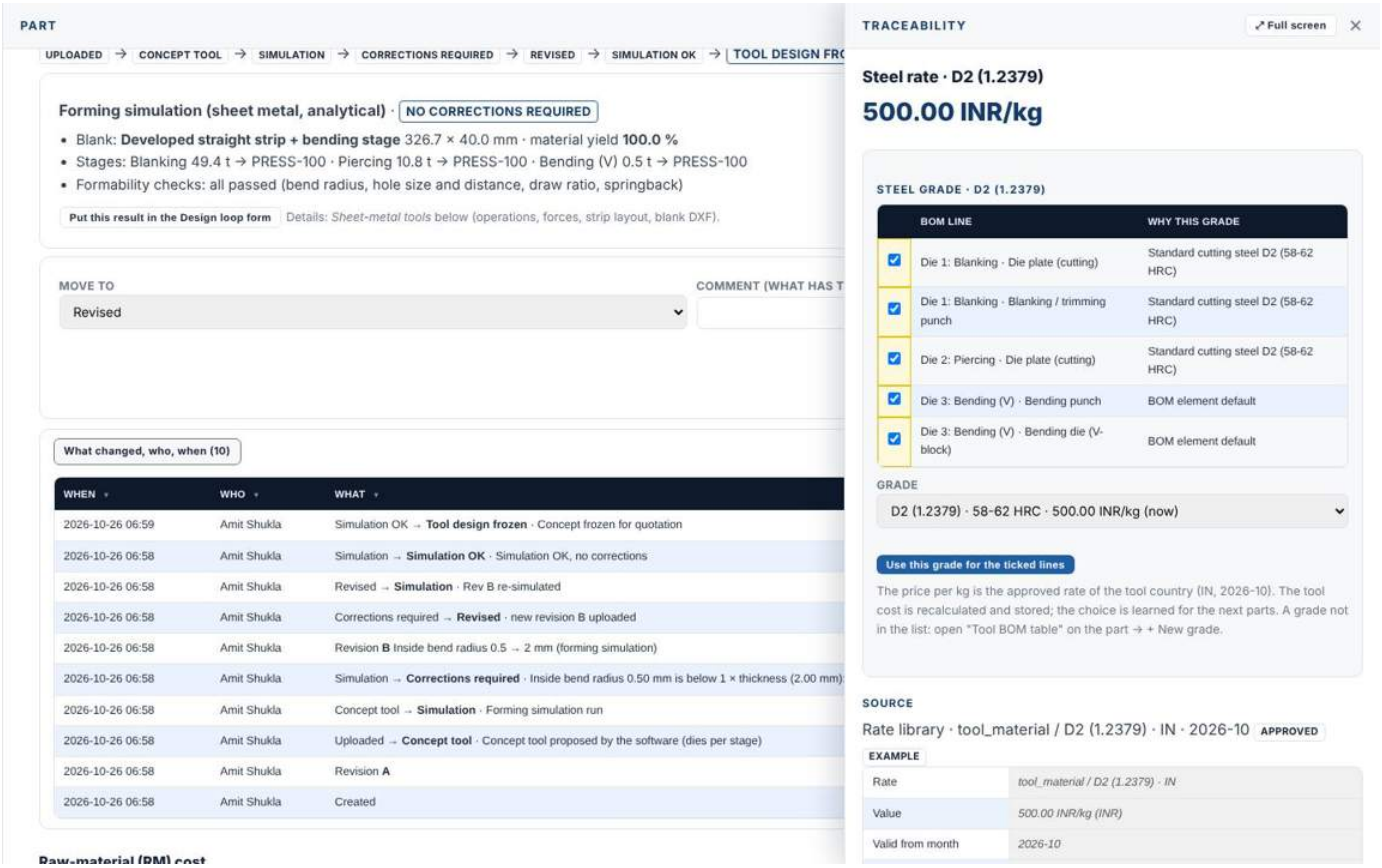


Traceability: click any value. Die set price: pick another die set for this part or add a row to master data.

Step 15 · Traceability: steel grade

• Click a steel rate (e.g. Steel rate · D2): why this grade (Grade rule / learned / your choice).
• Choose another grade (price per kg and % difference shown), tick the lines → Use this grade.

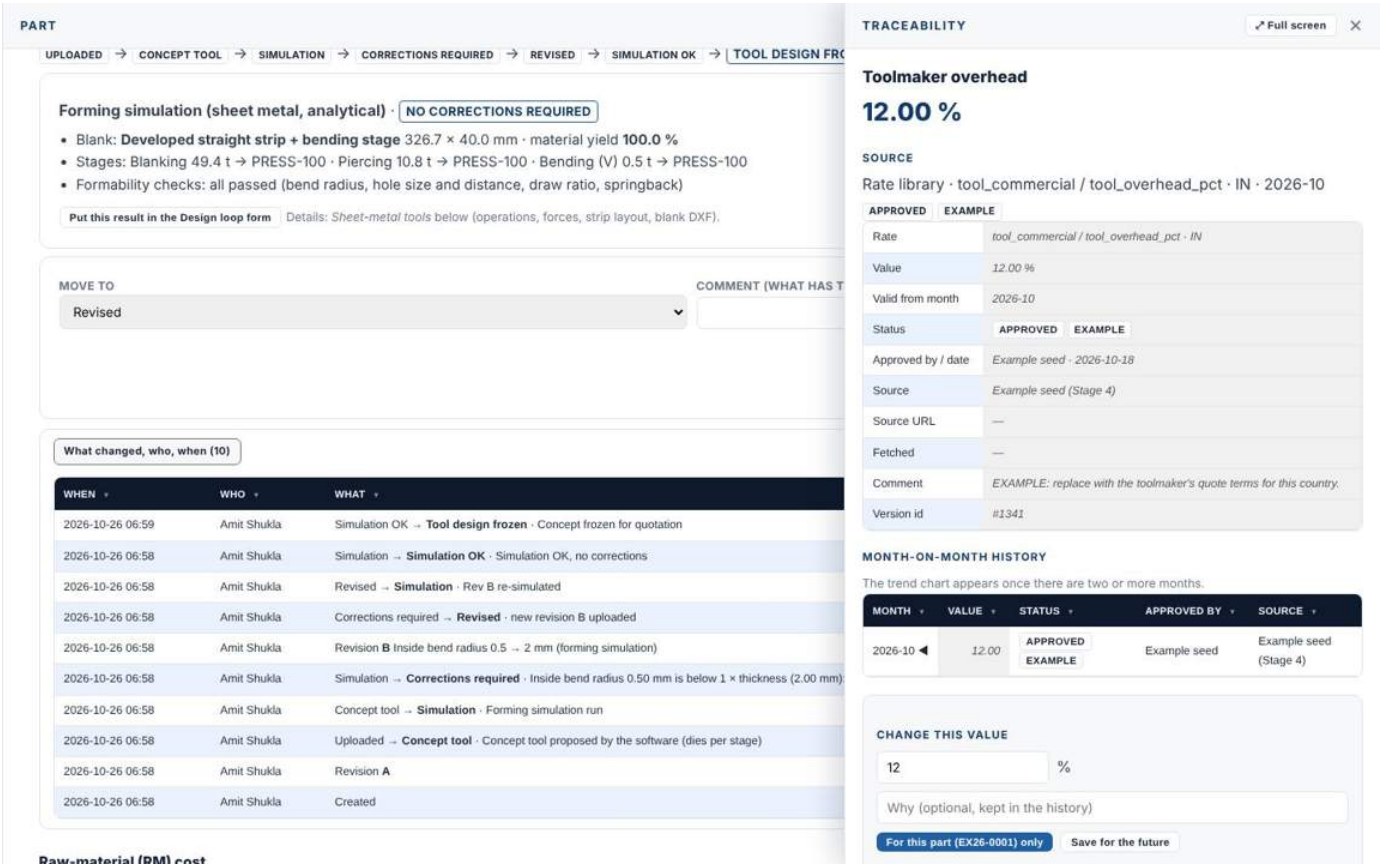
Output: New tool cost version; the choice is learned for the next parts.



Steel grade: why it was chosen, the grade drop-down with the price per kg, change it for the ticked lines.

Step 16 · Traceability: any value

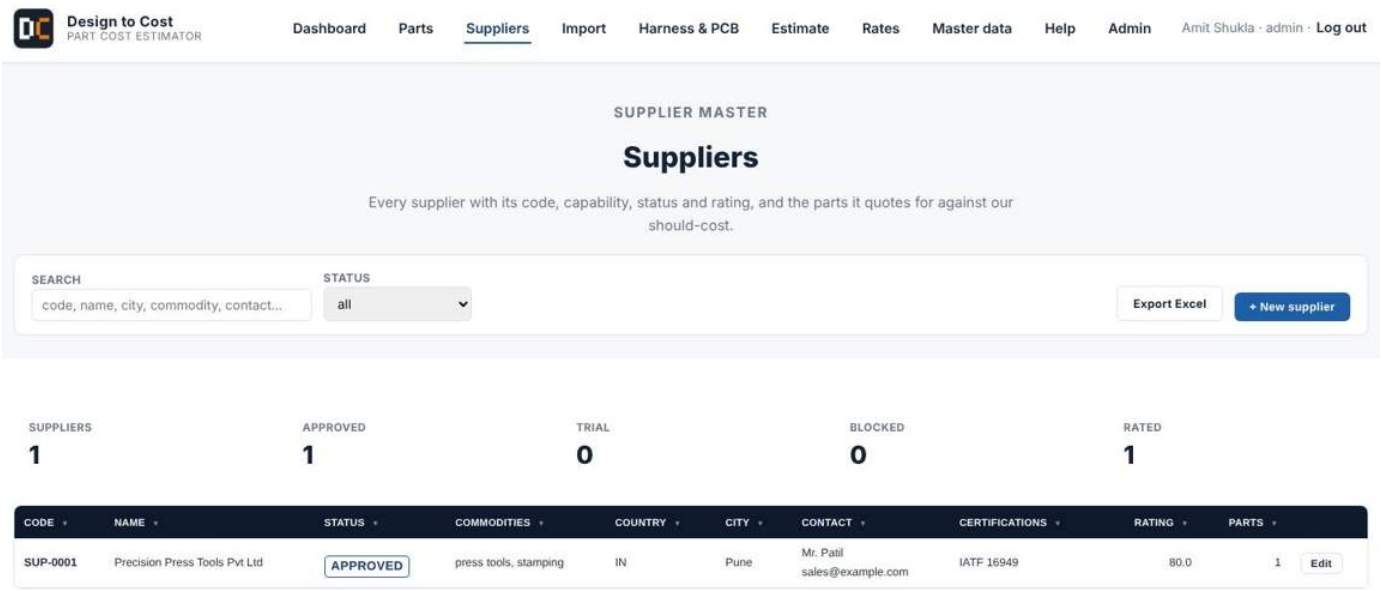
• Change this value: For this part only (this part's cost) or Save for the future (rate library / master data).
• Reset to master data removes your part-only change. Formulas can be changed the same way.



Any default value: “For this part only” or “Save for the future” (rate library / master data); reset any time.

Step 17 · Supplier master

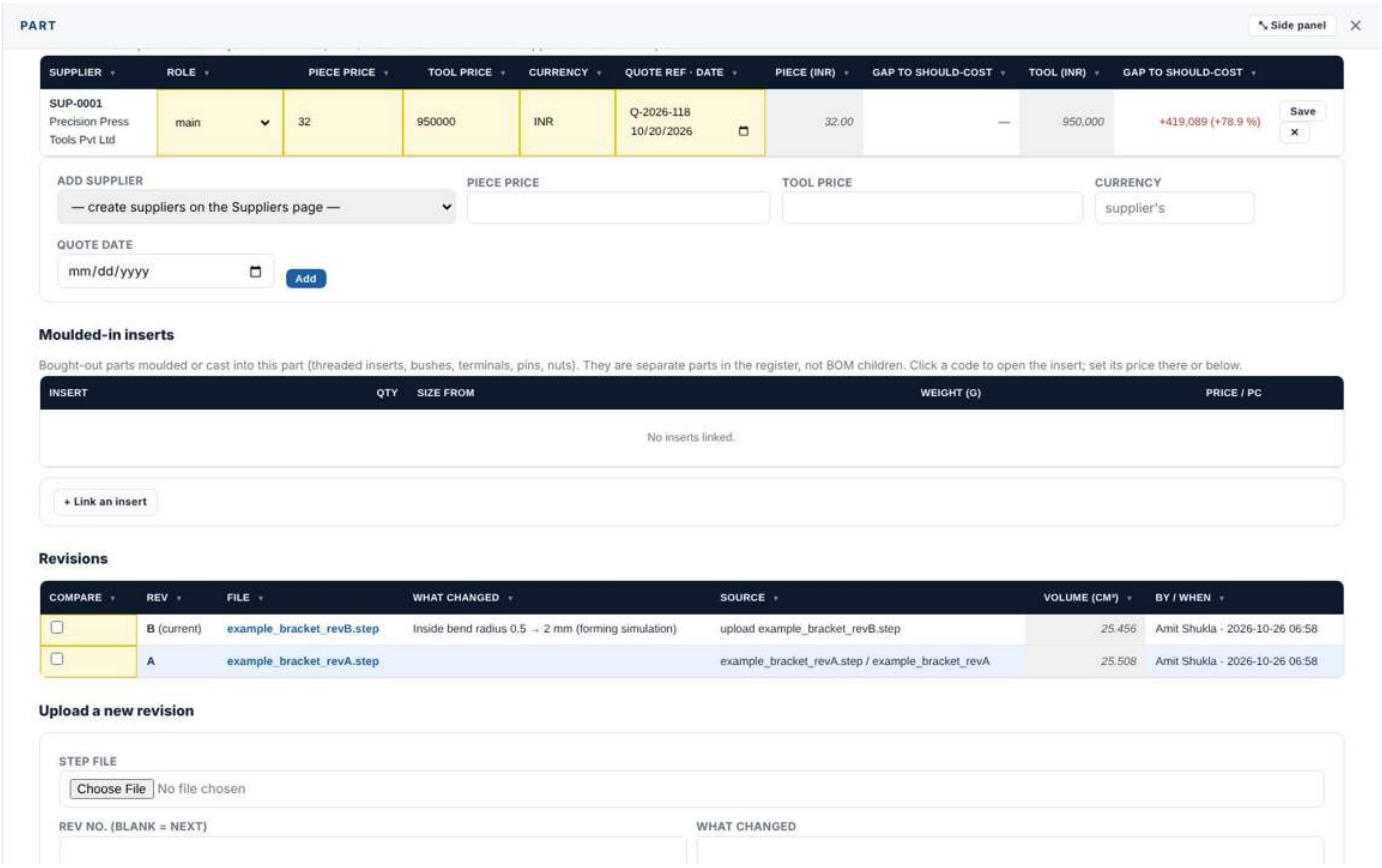
• Suppliers → + New supplier: code automatic (SUP-0001), status, capability, contacts, certifications, scores.
Output: suppliers.xlsx (Export Excel)



Suppliers: supplier master with automatic codes, status, capability and rating.

Step 18 · Supplier quotes against should-cost

• Part → Suppliers & quotes: add the supplier, piece price, tool price, quote ref and date.
• The gap to our should-cost is shown in INR and %.

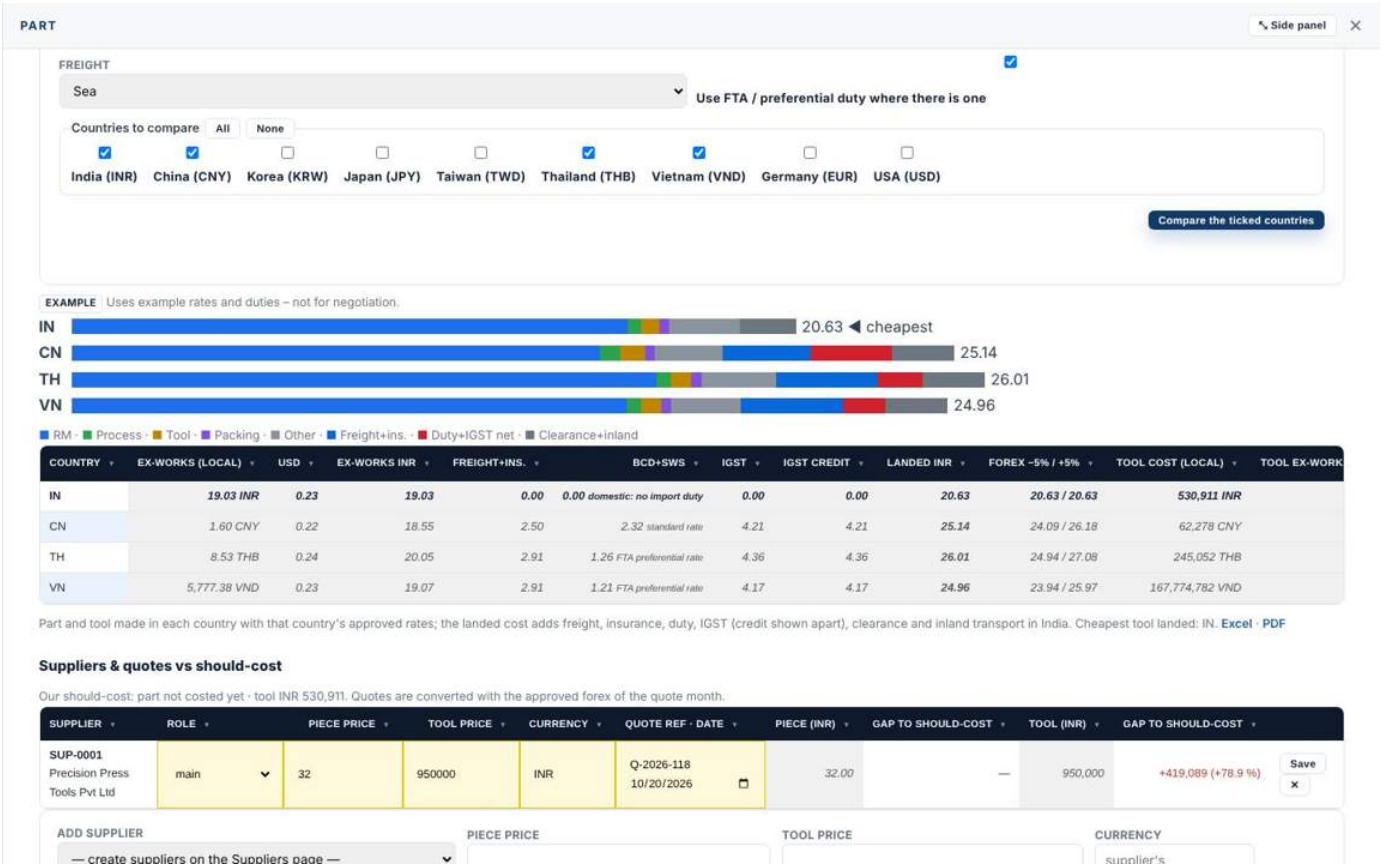


On the part: the supplier's quoted piece and tool price against our should-cost.

Step 19 · Compare countries

• Country comparison and landed cost: tick the countries (All / None) → Compare the ticked countries.
• Part and tool ex-works and landed in India; cheapest marked; Excel / PDF.

Output: country_comparison_….xlsx / .pdf

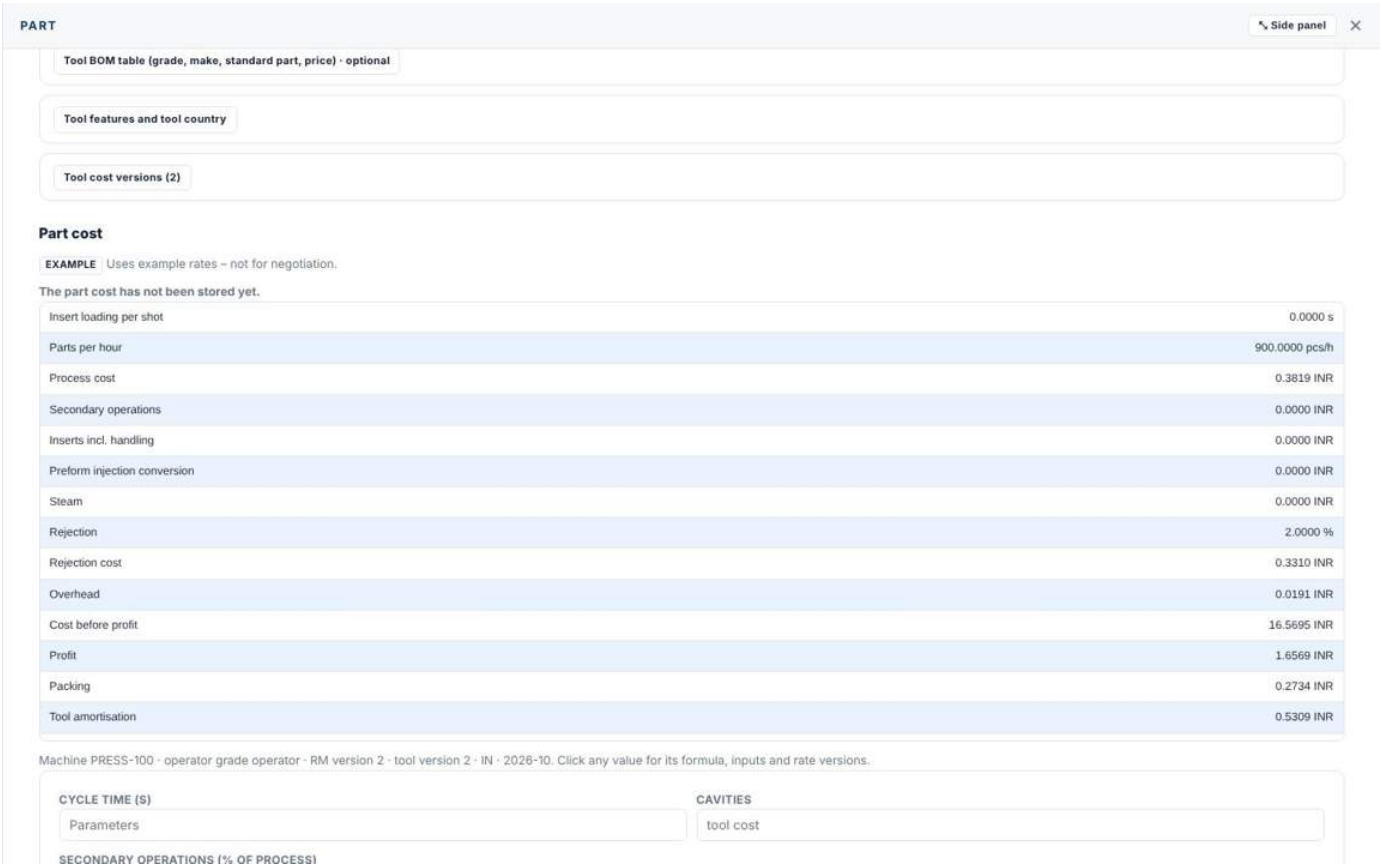


Country comparison: tick the countries; part and tool cost ex-works and landed in India, cheapest marked.

Step 20 · Download for RFQ / PO

• Tool cost → ■ PO pack (ZIP): part STEP and drawing, tool drawing and cost per stage, BOM per stage,
• blank / strip DXF, tool pack, RM and part cost. Send it with the RFQ; attach it to the PO.

Output: EX26-0001_PO_pack.zip

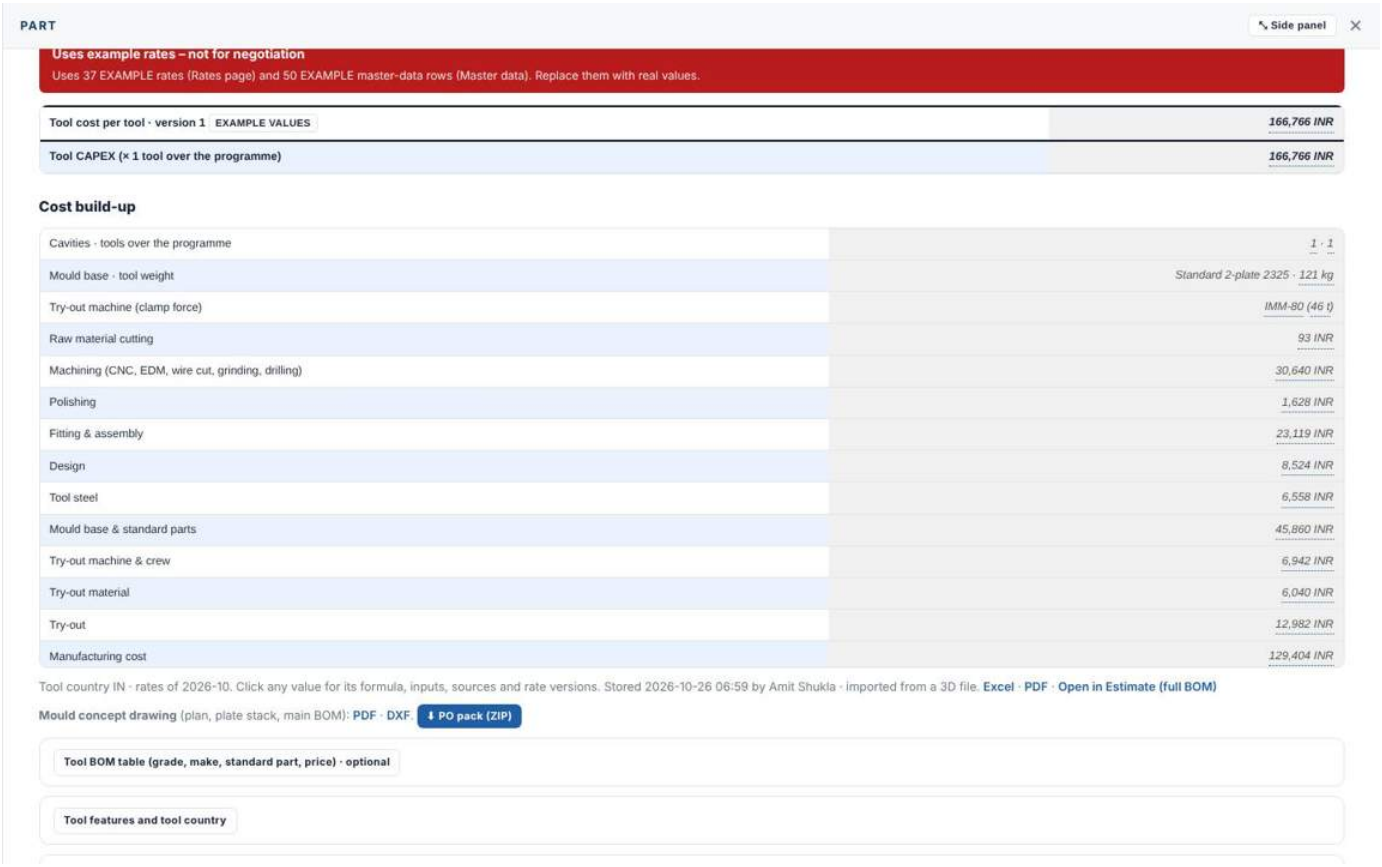


Downloads: tool drawings per stage, cost sheets and the PO pack (ZIP) with everything for the toolmaker.

Step 21 · A plastic part: the mould

• Import plastic_cover.step with material ABS: the mould is sized (cavities, base, inserts) and costed.
• Mould concept drawing PDF / DXF and the PO pack, the same way.

Output: EX26-0002_mould_drawing.pdf

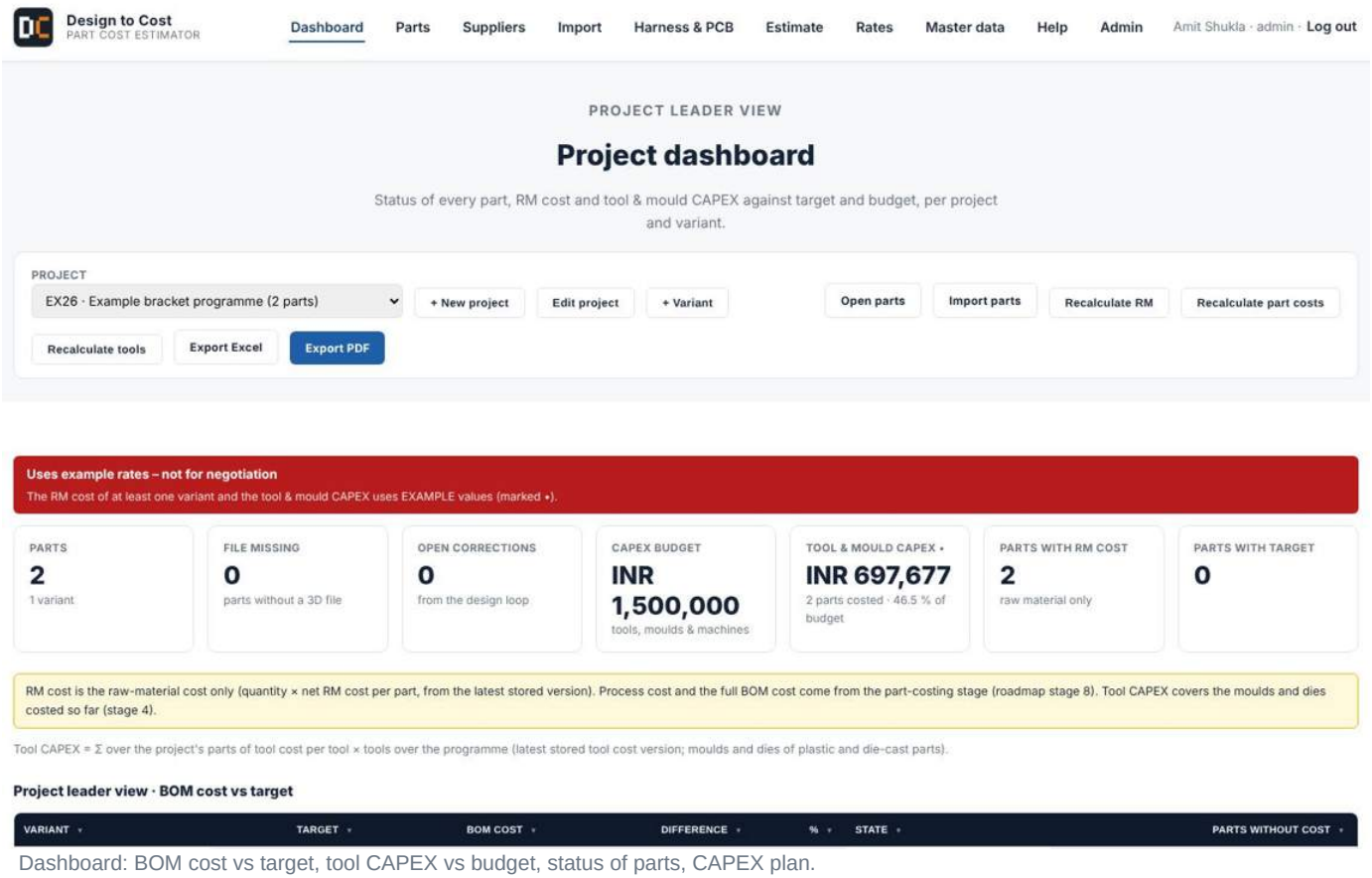


Plastic part: mould tool cost with mould base, steel grades from Grade rules, and the mould concept drawing.

Step 22 · Project view

- Dashboard: BOM cost vs target per variant, tool CAPEX vs budget, part status, CAPEX plan with suppliers.
- Recalculate RM / tools / part costs with the latest approved rates.

Output: Dashboard PDF, parts Excel

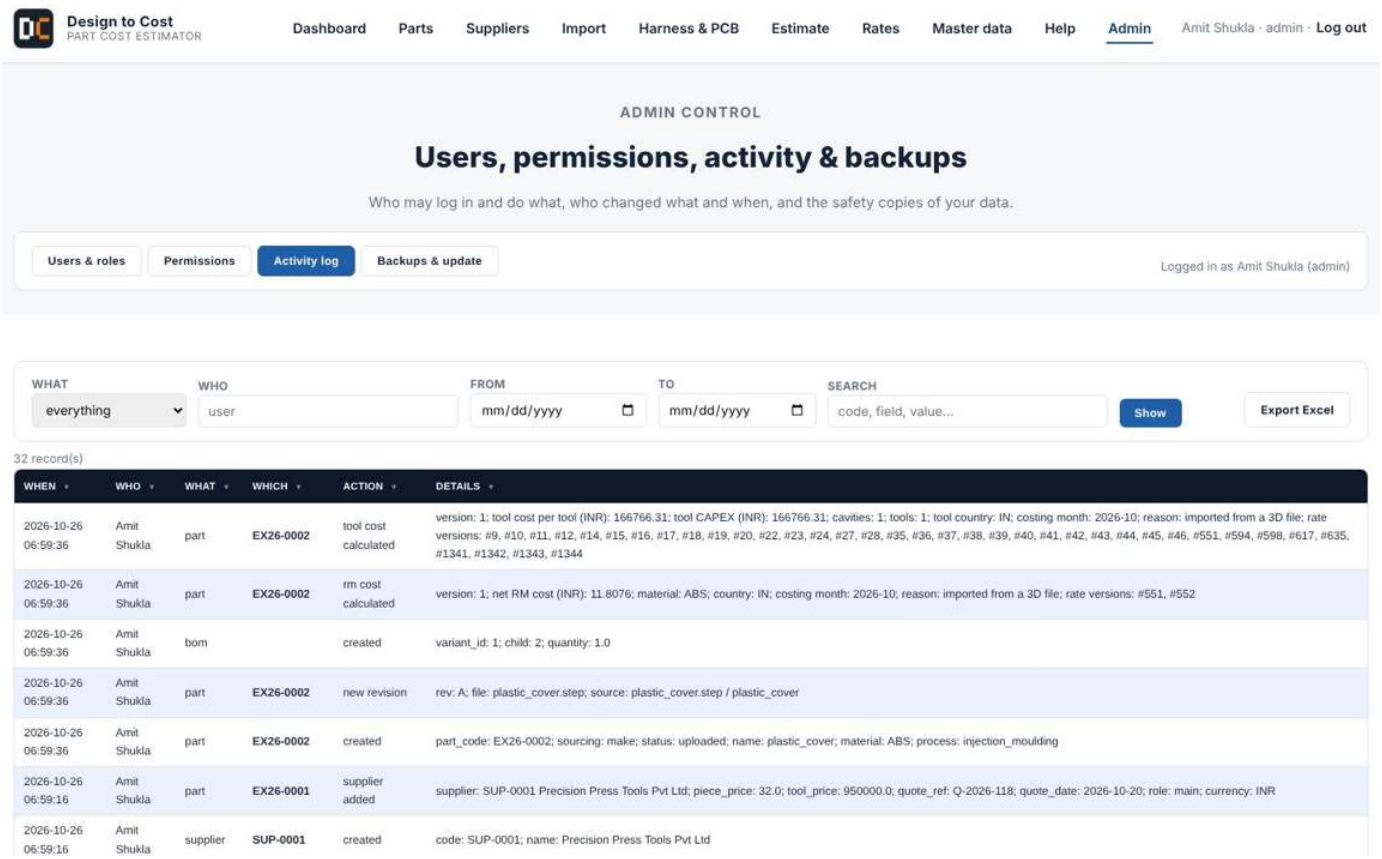


Dashboard: BOM cost vs target, tool CAPEX vs budget, status of parts, CAPEX plan.

Step 23 · Who changed what

• Admin → Activity log: filter by what / who / dates; Export Excel.

Output: activity_log.xlsx



Admin → Activity log: who changed what and when (Excel export).

3 · The design loop at a glance

Stage	What you do	What the software gives	Download
Uploaded	Import the STEP (Rev A).	Geometry, weight, RM, tool and part cost; process suggestion.	Part drawing PDF / DXF
Concept tool	Check the dies / mould proposed; change grades, base, operations if needed.	Dies per stage / mould with BOM and cost.	Tool drawing and cost per stage (PDF / DXF / Excel)
Simulation	Move to Simulation; read the result.	Sheet metal: forming result; plastic / die casting: wall, draft, sink, cycle, clamp.	Simulation PDF (plastic / die casting), tool pack
Corrections required	Put the result in the form → Change status; send the comment to the designer.	The correction kept in the timeline.	—
Revised	Upload the corrected model as a new revision.	All costs recalculated on the new revision.	Part drawing of the new revision
Simulation OK	Simulation again → Put this result → Simulation OK.	No corrections required.	—
Tool design frozen	Move to Tool design frozen.	Concept frozen for quotation.	PO pack (ZIP) for RFQ / PO

A step back (e.g. a forgotten Concept tool) is allowed with a comment: Move to → ■ Back to …

4 · Downloads for RFQ and purchase order

Tool cost → ■ **PO pack (ZIP)** holds everything the toolmaker needs; send it with the RFQ and attach it to the PO:

Folder / file	Content
README.txt	What is in the pack, and anything that could not be made
01_part / …step	The 3D model of the current revision (the master for all geometry)
01_part / …part_drawing.pdf / .dxf	Three views, overall dimensions, title block
02_tool / stageN_…_drawing.pdf / .dxf	Concept drawing of each die stage (or the mould): plan, plate stack, main BOM
02_tool / stageN_…_cost.pdf, …_bom.xlsx	Cost sheet and BOM (grade, make, part numbers) of each stage
02_tool / …tool_cost_all.pdf / .xlsx	All stages: summary and one page / sheet per stage
02_tool / …blank.dxf, strip.dxf, tool_pack.pdf	Blank, strip layout and stage sketches (sheet metal)
03_cost / …rm_cost.pdf, part_cost.pdf	Should-cost of material and of the part

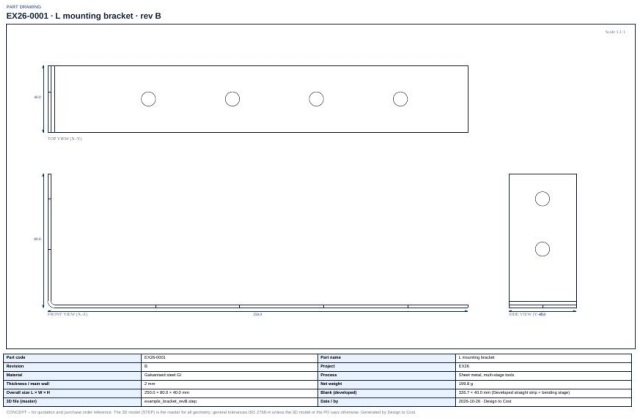
The drawings are CONCEPT layouts for quotation and PO reference; the 3D model is the master. Replace EXAMPLE rates and master-data values before using costs for negotiation.

5 · Corrections: what to change where

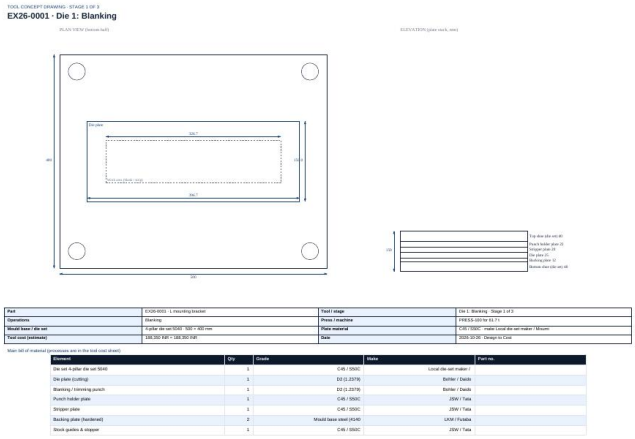
Nothing is lost when you correct: old revisions, cost versions and history stay; deletes make a backup first.

What is wrong	Where to correct it	Note
Part name, material, process, sourcing, volume, programme life	Parts → part → details form → Save changes	History keeps old → new
Wrong or changed 3D model	Part → Upload a new revision (old revisions stay)	Design loop → Revised
Part not needed in a project / not needed at all	Part → Remove from project BOM / Delete part (type the code)	Backup ZIP made first
Project data (budget, SOP, name) / whole project	Dashboard → Edit project / Delete project	Delete: type the code, backup first
Design-loop status or a forgotten step	Design loop → Move to → ■ Back to … with a comment	Shown as step back
Sheet-metal operations or die type	Sheet-metal tools → Operations in order (↑ ↓ ✕ + Add, Reset) / Die type → Save	Dies follow
Blank size, thickness, sheet size, gap	RM inputs on the part, or traceability → For this part only	Or Save for the future
Cavities, hot runner, slides, lifters, tool country	Tool cost → Tool features and tool country → Save and calculate	—
Steel grade of a BOM line or of the base	Traceability on the steel → grade drop-down → Use this grade	Learned for the next parts
Grade / make for every new tool	Tool BOM table → my default, or Master data → Grade rules / BOM elements	Master data
Mould base / die set	Traceability on the base price → Use for this part / + Add a row	Add row = master data
Standard part, make, quantity or price of a BOM line	Tool BOM table (optional) → change → Save; + New standard part	Part number kept
New steel grade	Tool BOM table → + New grade (price per kg, country, make)	Rate approved or draft
Any rate (material, scrap, steel, process, MHR input, labour, overhead %)	Traceability → For this part only / Save for the future; Rates page	New rate version
A rule or formula (BOM rule, cost formula)	Traceability on the formula → change → For this part only / Save for the future	Master data
Machines, countries, labour grades	Master data → Countries, machines & formulas	Approved rates on Rates
Supplier data or a quote	Suppliers page / Part → Suppliers & quotes	History kept
Users, roles, permissions	Admin → Users & roles / Permissions	Admin only
A stored cost looks old	Part → Recalculate, or Dashboard → Recalculate RM / tools / part costs	Old versions are never changed
Something wrong after an update	DTC_Rollback.bat (old version and data); DTC_Restore.bat (a backup)	Backups in DTC_Backups

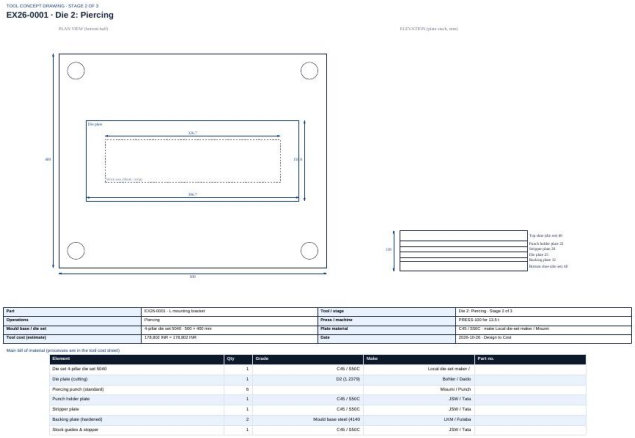
6 · The example outputs



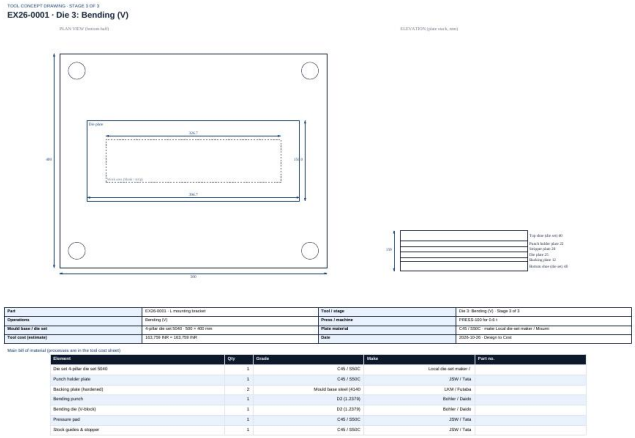
Part drawing (Rev B)
EX26-0001_part_drawing_revB.pdf



Stage 1 · blanking die drawing
EX26-0001_stage1_blank_tool_drawing.pdf



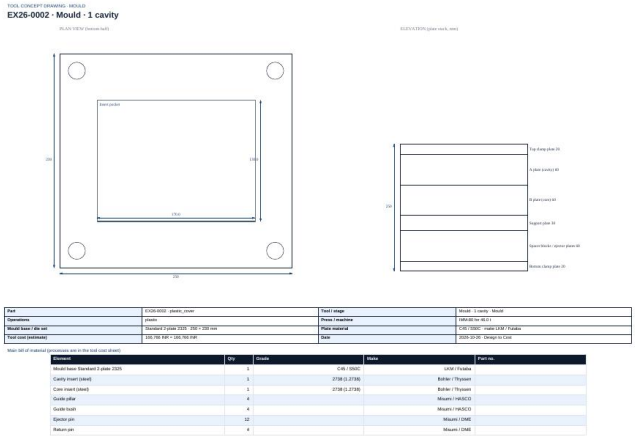
Stage 2 · piercing die drawing
EX26-0001_stage2_pierce_tool_drawing.pdf



Stage 3 · bending die drawing
EX26-0001_stage3_bend_tool_drawing.pdf



Tool cost, all stages
EX26-0001_tool_cost_all_stages.pdf



Mould concept drawing (plastic cover)
EX26-0002_mould_drawing.pdf

All files of this example: Help page → Work instruction → example outputs.

7 · Updates, backups and help

- Before an update nothing is needed: DTC_Update.bat backs up first. Check one known part after the update.
- Backups: DTC_Backups (copy them to OneDrive / a pen drive: backup_copy_to in dtc_settings.ini); Admin → Backups.
- Something wrong after an update: DTC_Rollback.bat. Lost data: DTC_Restore.bat.
- Help page: how every calculation works; README in the app folder: every feature in detail.